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# Continuidad de la norma

**Proposición 1.** *Sea $X$ un espacio normado, entonces la norma es continua.*

***Demostración*:** Sea $\varepsilon > 0$ y $x \in X$. Si tomamos $\delta = \varepsilon$, entonces, si $\|y - x\| < \delta$, por la desigualdad triangular inversa, se tiene que $|\|y\| - \|x\|| \leqslant \|y - x\| < \delta = \varepsilon$. ■

## Referenciado en

- `teo-cerrado-convexo-hilbert-imp-exists-min-norma`

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